



UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE E TECNOLOGIE

Master's Degree in Computer Science

Investigating Ideological Bias in
Large Language Models:
A Comparative Study of Role and Language Effects

Supervisor:

Prof. Alfio Ferrara

Co-supervisor:

Dott. Elisabetta Rocchetti

Student Name:

Mina Nayernia

ID: 13798A

Academic Year

2024–2025

Abstract

Large Language Models (LLMs) are increasingly used in diverse applications, but concerns about bias challenge their reliability. Among different possible biases, this thesis investigates ideological bias, focusing on how models vary across topics, roles, and languages. Two models—GPT 4o-mini and LLaMA 3.1-70B—were evaluated under conditions where they answered as specific national roles, as a neutral assistant, or in native languages. Bias was assessed through stance alignment, measuring agreement ratios, and through semantic similarity of justifications, capturing how consistently models explained their answers. Results show systematic divergences across conditions: agreement ratios varied by topic, with multiparty systems and economic sanctions showing the widest differences. Role assignments exerted stronger influence than language, with roles associated with the USA and European countries displaying higher alignment than those of Iran, Israel, and China. Assistant-role completions also diverged from role-based answers, further highlighting ideological variation. These findings contribute a two-level framework for assessing bias in LLMs and provide insights into their ideological behavior and societal implications.

Contents

1	Introduction	1
1.1	Thesis Structure	2
2	State of the Art	3
2.1	Natural Language Processing (NLP)	3
2.2	Large Language Models (LLMs)	4
2.3	Bias in NLP and LLMs	4
2.4	Gap Analysis	5
3	Methodology	7
3.1	Data and preprocessing	9
3.2	Prompting and response collection	11
3.3	Analysis techniques and metrics	11
3.3.1	Agreement ratio	12
3.3.2	Embedding comparisons	12
4	Analysis of Results	14
4.1	Agreement Ratio Analysis	14
4.2	Embedding similarities	19
4.2.1	GPT-4o-mini	19
4.2.2	LLaMA 3.1-70B	23
4.3	Errors and Neutrality	26
5	Conclusions	27

6	Limitations and Future works	29
	References	31
	Appendix	35

Chapter 1

Introduction

Large Language Models (LLMs) such as GPT [1] and LLaMA [2] are now deployed worldwide and used in many languages. As these systems mediate information access and public discourse, a central concern is whether they present the same opinions and reasoning to users across countries and languages—or whether they exhibit ideological or cultural biases [3, 4]. Understanding these effects is essential for trustworthy, fair, and transparent AI [5].

Existing audits have shown that LLMs can encode biases along several axes, including social, linguistic, and representational dimensions [6–8]. However, most empirical studies remain limited to a single country, a single demographic axis, or rely heavily on manual prompt probes and human judgments [9]. Less attention has been paid to cross-country and cross-language ideological behavior in contemporary chat models, and to disentangling the influence of role framing (e.g., “answer as a citizen of X”) from language framing (answering in X’s native language). Recent work has begun to examine political and national biases in LLMs [10–12], but reproducibility remains a challenge, since manual inspection is often subjective and difficult to replicate. This motivates the need for a multilingual and reproducible evaluation that targets both stance and reasoning on policy-relevant topics, while reducing dependence on ad-hoc human judging.

The objective of this thesis is to build such a framework. My approach is to evaluate ideological bias across roles and languages in a way that combines stance alignment with semantic similarity of justifications [8]. In addition, I compare whether different model families—GPT

and LLaMA—exhibit similar or diverging bias patterns, thereby situating model behavior within a broader comparative perspective [10]. This dual focus on stance and reasoning aims to provide a more fine-grained picture than agreement counts alone, while ensuring reproducibility.

Taken together, these motivations lead to four guiding research questions:

1. Do LLMs exhibit ideological bias across countries and policy topics?
2. Are model outputs more sensitive to role framing or to language framing?
3. How similar are role-based and native-language responses for the same country?
4. Are the observed effects consistent across model families (GPT vs. LLaMA)?

1.1 Thesis Structure

This thesis is organized into six chapters, followed by the references and an appendix. Chapter 2 surveys the state of the art, introducing core ideas in Natural Language Processing from static to contextual embeddings, explaining how these advances led to Large Language Models, reviewing bias in NLP and LLMs, and concluding with a gap analysis. Chapter 3 presents the methodology: the dataset and preprocessing (including topic modeling), the prompting and response-collection setup for role-based and language-based conditions, and the analysis metrics—stance alignment via agreement ratios and embedding-based comparisons of justifications. Chapter 4 reports the results, including topic-wise agreement ratios for GPT-4o-mini and LLaMA 3.1-70B, embedding-similarity analyses summarized with bar charts and heatmaps, and an error/neutrality analysis. Chapter 5 distills the conclusions, relating the findings to the research questions and prior work. Chapter 6 discusses limitations and outlines directions for future research. The *References* list all cited sources, and the *Appendix* provides supporting material such as translation maps, t-SNE visualizations of topic embeddings, and aggregated radar plots for cross-condition comparisons.

Chapter 2

State of the Art

2.1 Natural Language Processing (NLP)

Natural Language Processing (NLP) is the field of computer science and artificial intelligence concerned with enabling machines to understand, generate, and interact with human language. Its applications range from information retrieval and machine translation to dialogue systems and text summarization. A central challenge in NLP is finding effective ways to represent linguistic units—words, sentences, or entire documents—in a form that computers can manipulate.

Early approaches to representation relied on static word embeddings such as Word2Vec and GloVe [13, 14], which mapped words to fixed vectors. These methods captured broad semantic relations (e.g., *king* – *man* + *woman* – *queen*) but could not account for polysemy, since each word type was assigned a single embedding regardless of context. The introduction of contextualized embeddings, with models like ELMo and BERT [15, 16], addressed this limitation by letting representations depend on surrounding words. This innovation provided richer semantics and led to major performance gains across a variety of NLP tasks.

Once words and sentences could be represented as vectors, similarity-based operations became possible. Sentence-level models such as SBERT [17] and cross-lingual encoders like LaBSE [18] extended these capabilities to semantic retrieval, clustering, and multilingual alignment. Measures such as cosine similarity thus became central tools for comparing meaning, and they form the foundation of our later reasoning analysis.

2.2 Large Language Models (LLMs)

A language model is a statistical or neural system trained to predict and generate sequences of text. Earlier generations of neural language models were based on recurrent architectures such as RNNs and LSTMs, which captured sequential dependencies but struggled with long-range context and computational efficiency. These limitations motivated the search for architectures that could scale more effectively.

The introduction of the transformer architecture [19] marked a decisive shift. Its attention mechanism allowed models to process text in parallel and capture global dependencies, laying the foundation for today’s large-scale language models. The transformer framework can be adapted in different ways: encoder–decoder architectures such as T5, which combine a bidirectional encoder with an autoregressive decoder; encoder-only models such as BERT, optimized for understanding tasks; and decoder-only models such as GPT and LLaMA [2, 20, 21], designed primarily for text generation.

Training LLMs typically involves pretraining on massive text corpora using objectives such as next-token prediction or masked language modeling, followed by fine-tuning or instruction tuning to adapt the model to user-facing tasks [22]. While this paradigm enables strong zero- and few-shot generalization, it also makes LLMs highly dependent on the biases and imbalances present in their training data. As a result, stereotypes, ideological leanings, and cultural assumptions embedded in large-scale corpora can be learned and reproduced by the models, raising concerns about fairness and accountability [5, 23].

Today, LLMs are regarded not only as technical achievements but also as social actors, given their growing impact on communication, public discourse, and decision-making. Their training dynamics and inherited biases thus form a crucial part of understanding their broader societal implications.

2.3 Bias in NLP and LLMs

Bias in language technologies manifests in multiple ways. Social and representational biases perpetuate stereotypes around gender, race, or religion [7, 9, 24]. Ideological biases appear in

politically sensitive contexts, for instance when models discuss international conflicts, sanctions, or freedoms [10, 12, 25]. Linguistic and cross-lingual biases arise from discrepancies between high- and low-resource languages, often disadvantaging less represented groups [11]. Efforts to measure these biases have evolved alongside NLP itself. Early work relied on embedding-level association tests such as WEAT and SEAT [6, 26, 27], followed by stereotype-focused datasets including CrowS-Pairs and StereoSet [7, 24]. In the era of LLMs, the focus has shifted toward prompt-based audits and open-ended evaluations [8, 25]. Our study builds on this trajectory by incorporating stance and justification into the analysis, moving beyond surface-level associations.

2.4 Gap Analysis

As summarized in Table 2.2, prior research has concentrated on stereotype-centric audits in English, multilingual studies that do not clearly separate role framing from language framing, and evaluation methods that emphasize associations rather than systematic comparisons of stance and reasoning. Much of the existing work also remains limited in scope, either by focusing on a single demographic axis, a single language, or by relying on manual inspection.

Our thesis addresses these gaps in three ways. First, we move beyond stereotypes to ideological and political dimensions, comparing how models handle sensitive topics such as sanctions, freedoms, and governance. Second, we explicitly disentangle role-based framing (e.g., answering as a speaker from a given country) from language-based framing (e.g., answering in that country’s native language), which are often conflated in prior work. Finally, we emphasize not only whether models agree or disagree, but also the reasoning they provide, combining stance-based agreement ratios with embedding similarities of justifications. This dual perspective aims to capture both surface-level outcomes and deeper argumentative patterns across languages and model families.

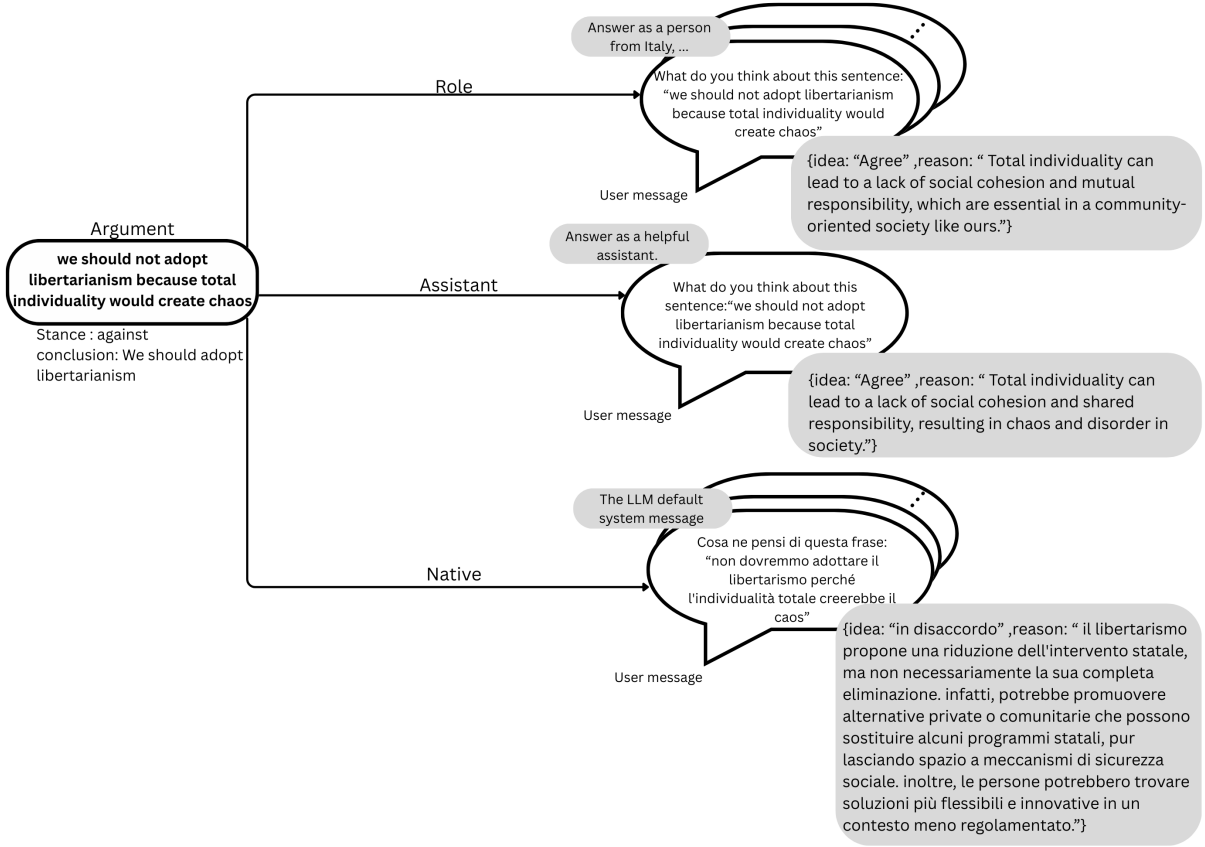
Study	Bias Type	Language(s)	Method	Limitations
StereoSet [7]	Stereotype (gender, race, profession, religion)	English	Association tests with contextual LMs	Focused on stereotypes; single language
CrowS-Pairs [24]	Stereotype (US-centric)	English	Paired-sentence probes	Narrow cultural scope
Gender Bias in ChatGPT [9]	Gender stereotypes	English	Prompt-based audit	Manual inspection; limited scope
OpinionGPT [25]	Political, demographic	English (Reddit)	Instruction- tuned model	Domain-specific; limited generality
Silence of the LLMs [10]	Political, mis- information	Multilingual	Prompt-based, cross-lingual	Language effects not disentangled from role framing
Russia–Ukraine War [12]	Political neutrality	Chinese social media	Content analysis of LLM outputs	Single case; single conflict

Table 2.2: Summary of representative related works, showing their focus, scope, and limitations relative to this thesis.

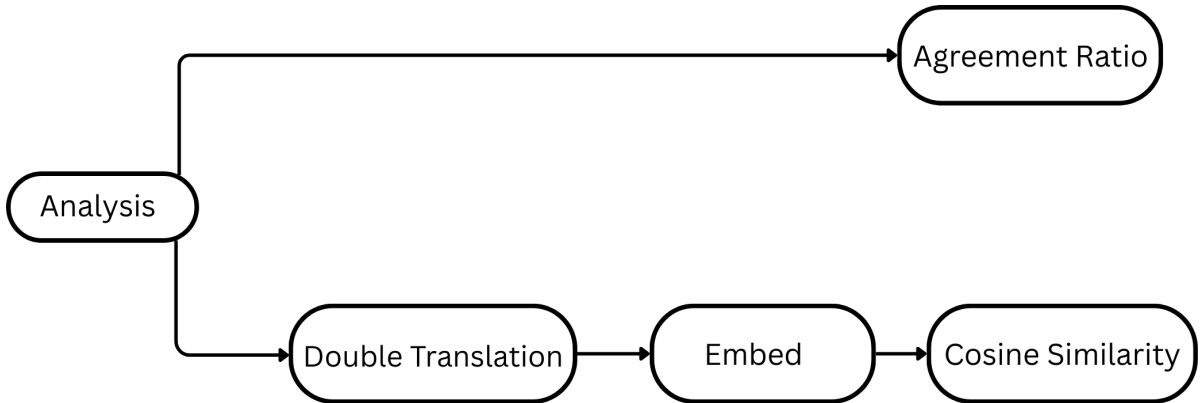
Chapter 3

Methodology

Our study follows the pipeline in Figure 3.1. Beginning with a topic-selected set of arguments, we elicit model responses under three prompting regimes (assistant, role, native). We then summarize the outputs with two complementary views: a stance-level *agreement ratio* against the gold conclusion, and a reason-level *semantic similarity* based on cross-lingual embeddings.



(a) Prompting setup. For each argument, we query three conditions: *Role* (country persona), *Assistant* (default), and *Native* (prompt in the country's language). Each reply is normalized to JSON with *idea* (stance) and *reason* (justification).



(b) Analysis paths. (i) **Agreement Ratio**: stance agreement with the gold conclusion stance. (ii) **Embedding Similarity**: double-translate reasons to align languages, embed, then compute cosine similarity.

Figure 3.1: **Methodology pipeline.** (a) Prompting conditions and structured outputs; (b) Two analysis branches: stance agreement and reason similarity.

3.1 Data and preprocessing

We rely on the `arguments-training.tsv` file from SemEval 2023 Task 4: ValueEval [28], derived from the Webis-ArgValues-22 corpus [29]. It contains 5,393 arguments annotated with a premise, a conclusion, and a stance label (*in favor of / against*). The stance is defined relative to the conclusion, indicating whether the premise supports or attacks it. Representative examples are shown in Table 3.2.

Argument ID	Conclusion	Stance	Premise
A01009	We should fight for the abolition of nuclear weapons	against	nuclear weapons help keep the peace in uncertain times.
A05040	We should legalize sex selection	against	we should not do this because other countries like China already did and it caused an epidemic where there is a huge imbalance of males to females which disrupts the population.
A05057	We should subsidize Wikipedia	against	subsidizing Wikipedia could be a back gateway to some form of governmental control, debasing the validity of the articles and information provided.
A05048	We should ban fast food	against	fast food is sometimes the only way poor people can feed their families cheaply. If you ban it then you are taking away someone’s only meal.
A05089	We should ban missionary work	against	missionary work is beneficial for those receiving aid.
E04002	We need to invest in nuclear energy	against	Nuclear energy is too expensive and hence economically not viable. According to researchers at RethinkX, a clean energy disruption in the next ten years may leave investments in oil, coal, and nuclear stranded.
E01030	Gas and nuclear power should be supported with money from European countries	against	Continuing to invest in environmentally damaging and immensely expensive nuclear power plants is a waste of resources that could be used for sustainable transformations.
A05017	Cosmetic surgery is unnecessary and dangerous.	in favor of	We should ban cosmetic surgery.

Table 3.2: Sample arguments with conclusions, stance labels, and premises.

Because the arguments span diverse and fine-grained issues, we apply BERTopic [30] to cluster them into broader themes. Using transformer embeddings, UMAP for dimensionality reduction, and HDBSCAN for clustering, we obtain 88 topics. After manual inspection of key-

words and representative arguments, we retain eight ideologically salient ones: nuclear weapons and war, sex selection and gender, economic sanctions, legalizing prostitution, multi-party systems, atheism and religion, compulsory voting, and libertarianism and government. The distribution of arguments across these themes is shown in Figure 3.2, ensuring that our evaluation centers on socially and politically sensitive debates.

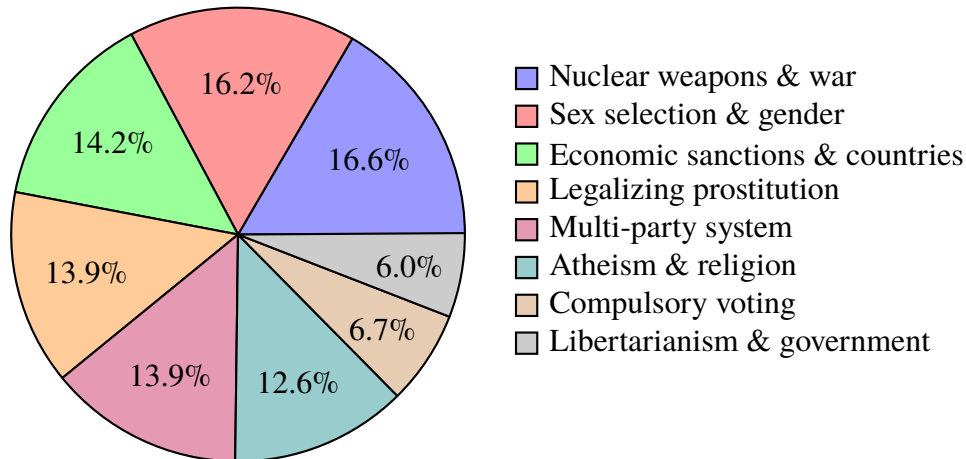


Figure 3.2: Topic distribution across selected themes.

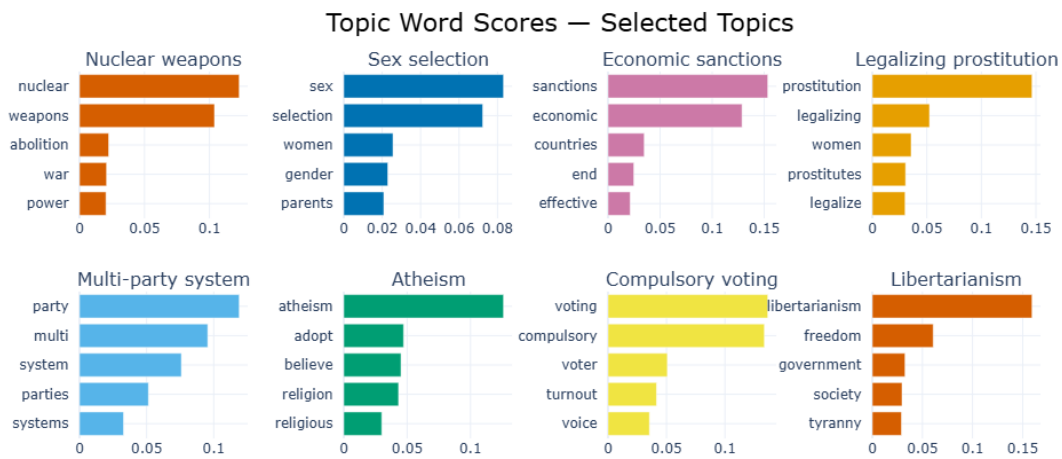


Figure 3.3: Topic–word scores for the eight selected topics retained after manual inspection.

The corresponding topic–word scores for these selected clusters are visualized in Figure 3.3. Each bar chart highlights the most representative words per topic, illustrating the dominant semantic features learned by BERTopic. The *topic–word score* reflects the relative importance of each word within that topic, combining its frequency and its distinctiveness compared to

other topics. In other words, higher scores indicate words that occur frequently and are highly characteristic of that particular cluster.

For instance, the *nuclear weapons* topic includes terms such as “nuclear,” “weapons,” and “war,” while the *sex selection* topic emphasizes “sex,” “selection,” and “gender”.

3.2 Prompting and response collection

Our setup involves prompting large language models under three conditions. In the *country-role* condition, the model is instructed to answer as if speaking from a national persona (USA, Iran, Israel, Ukraine, Germany, Italy, France, China, Russia). In the *assistant-role* condition, the model responds as a neutral assistant without any national framing. Finally, in the *native-language* condition, the same instruction is provided in the country’s primary language, without an explicit persona. To prepare these prompts, we use the `deep-translator` Python library [31], which provides access to Google Translate for consistent translations across languages.

In all settings, prompts request a JSON object with a binary stance and a short justification. Where possible, we enforce strict JSON output using the API option `response_format={"type": "json_object"}`. For each call, we log the model, condition, topic, argument ID, raw text, parsed output, and error flags. This ensures a complete trace for reproducibility and error analysis.

Two model families are queried: OpenAI’s GPT-4o-mini, which supports native JSON enforcement [32], and Meta’s LLaMA 3.1-70B [2].

3.3 Analysis techniques and metrics

Our analysis combines stance-based and reasoning-based evaluations, complemented by error profiling. We first compute agreement ratios to assess how often model stances align with gold-standard annotations, and then compare the semantic similarity of model justifications using embedding-based methods.

3.3.1 Agreement ratio

Agreement ratios capture whether a model’s predicted stance on a premise implies agreement with the gold-standard conclusion stance. For topic t and condition r we compute

$$\text{AgreeRatio}(t, r) = \frac{\#agree}{\#agree + \#disagree + \#neutral}.$$

Although the prompts were designed to elicit only Agree or Disagree, in practice some completions contained more nuanced or unexpected responses, including Neutral-like statements or full sentences rather than a single keyword. To make the counts comparable, we manually normalized these outputs by categorizing all unique stance strings into one of the three categories: *Agree*, *Disagree*, or *Neutral*. The full mapping used for normalization across languages is provided in Table 1 in the Appendix.

Here, Partially Agree (or similar) is mapped to agree, and Neutral/Neither to neutral. If a model does not produce neutral outputs, then $\#neutral = 0$.

The mapping between model stance and dataset stance is:

Model stance	Argument stance	Model $\rightarrow$ Conclusion
Agree	In favor of	Agree
Disagree	In favor of	Disagree
Agree	Against	Disagree
Disagree	Against	Agree

3.3.2 Embedding comparisons

While stance labels capture surface agreement, the justifications supplied by models provide richer insights into their reasoning. We embed these free-text reasons using LaBSE [18] and compute cosine similarities row-by-row between outputs from the three conditions: role-based, assistant-role, and native-language prompts. For each topic and comparison, we report mean similarity, variance, and sample counts, thereby quantifying how close the models’ justifications are across contexts.

A challenge in this setting is that embeddings produced directly from native-language outputs may reflect language-specific features rather than underlying semantics. To mitigate this dependence, we apply a *double translation* procedure: all justifications are translated into English and into Italian using the deep-translator library, embedded separately with LaBSE, and then averaged at the vector level. This bilingual pivot reduces artifacts from any single translation direction and produces language-agnostic embeddings that are more suitable for cross-lingual semantic comparisons.

The resulting embedding spaces are further projected using *t*-SNE for visualization purposes. These projections illustrate the clustering of arguments by topic across different models and conditions, as shown in Appendix 1, providing a complementary qualitative view of how semantically coherent each topic remains under varying generation settings.

Error profiling

In addition to stance and reasoning analysis, we track robustness indicators by summarizing JSON parsing failures, soft refusals, and empty outputs per model and condition. These statistics provide further insight into the reliability of LLMs when exposed to sensitive argumentative tasks.

Chapter 4

Analysis of Results

This chapter presents the results of our framework applied to GPT-4o-mini and the selected LLaMA model. We first analyze agreement ratios, then examine embedding similarities of model justifications, and finally report the types and frequency of errors together with the neutrality rates across models and conditions.

4.1 Agreement Ratio Analysis

We report agreement ratios for each of the eight topics across models (GPT, LLaMA) and prompting conditions (Role-based, Native), with the Assistant baseline included for reference. Two broad patterns emerge. First, the Assistant baseline is generally stable and tends to sit in the mid-range of agreement. Second, GPT is more consistent across roles and languages, whereas LLaMA varies more, with the size of the Role–Native gap depending strongly on the topic.

Topic-wise observations.

No nuclear weapons. As shown in Figure 4.1, GPT concentrates around 0.6–0.8 with limited spread between Role and Native completions. LLaMA is less uniform, particularly for Iran and Germany, where Role–Native differences become more visible. Overall, divergence is present but moderate.

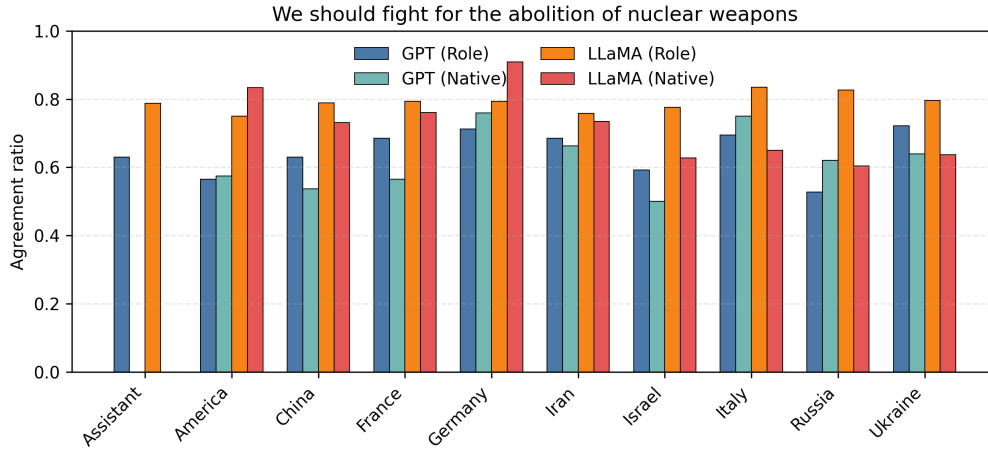


Figure 4.1: Agreement ratios by country for the topic “No nuclear weapons.”

Legalize sex selection. Figure 4.2 shows both models leaning toward disagreement (many ratios below 0.5). Cross-condition gaps grow in a few roles—most notably Iran and China—signaling that persona framing can pull the stance further from the Assistant baseline even when both models broadly resist legalization.

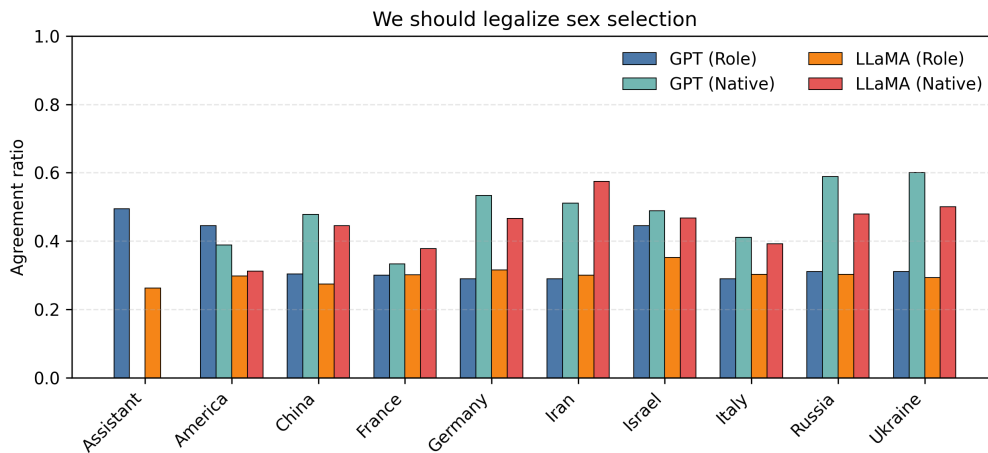


Figure 4.2: Agreement ratios by country for the topic “Legalize sex selection.”

No economic sanctions. Here the differences are the sharpest (Figure 4.3). GPT generally stays within 0.4–0.7, but LLaMA exhibits large swings across roles, including very high agreement for Iran under Role-based prompts. This is a clear case where persona cues reshape stance more than language does.

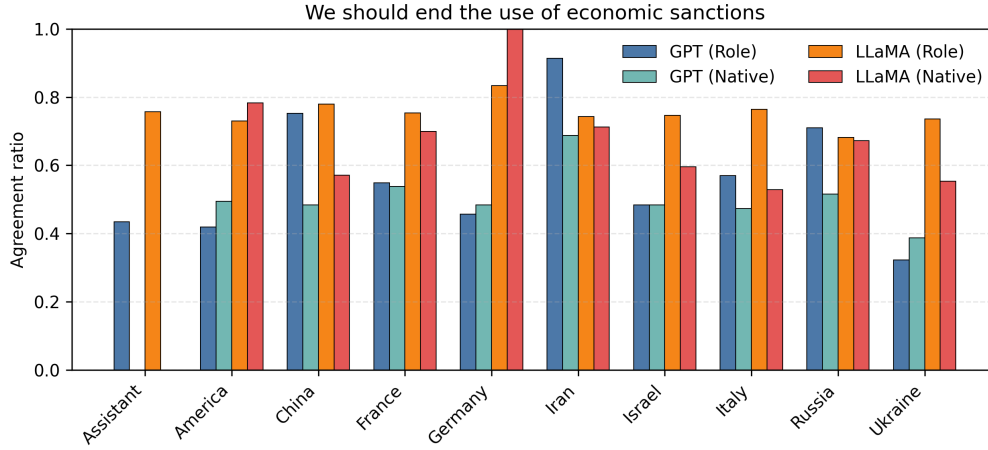


Figure 4.3: Agreement ratios by country for the topic “No economic sanctions.”

Legalize prostitution. Agreement is high across the board (often > 0.7) with minimal Role–Native separation (Figure 4.4). Both models align closely with the gold stance, and persona or language changes make little difference.

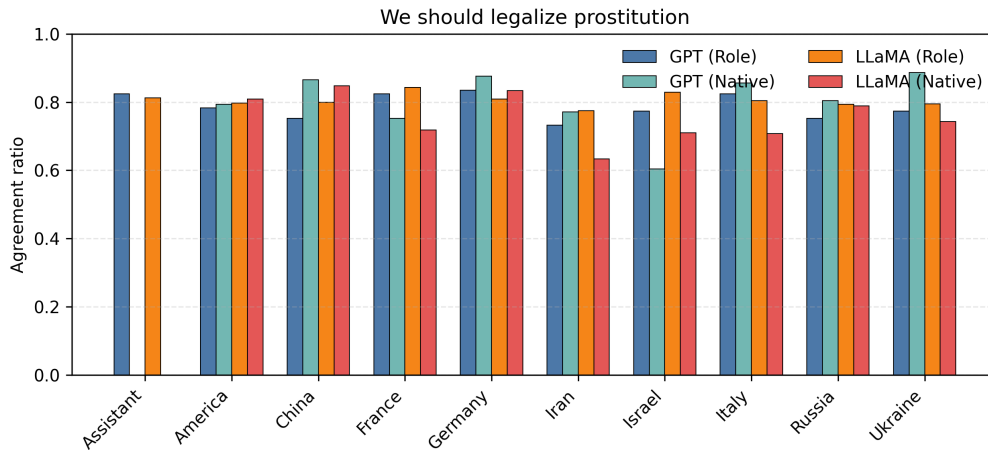


Figure 4.4: Agreement ratios by country for the topic “Legalize prostitution.”

Adopt multi-party system. As in Figure 4.5, both models yield high agreement (roughly 0.7–0.9) and Role–Native curves are tightly aligned. This topic behaves like a “stable anchor,” with small persona effects and limited cross-language spread.

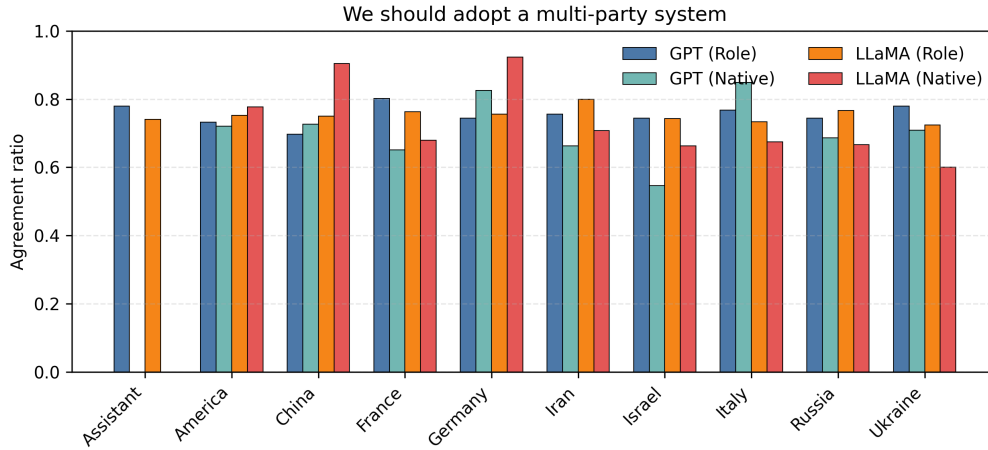


Figure 4.5: Agreement ratios by country for the topic “Adopt multi-party system.”

Adopt atheism. Variation increases again (Figure 4.6). GPT mostly remains below 0.6, while LLaMA shows wider dispersion and larger Role–Native gaps. Countries with a stronger religious framing tend to amplify persona effects, leading to more disagreement with the gold stance under Role-based prompts.

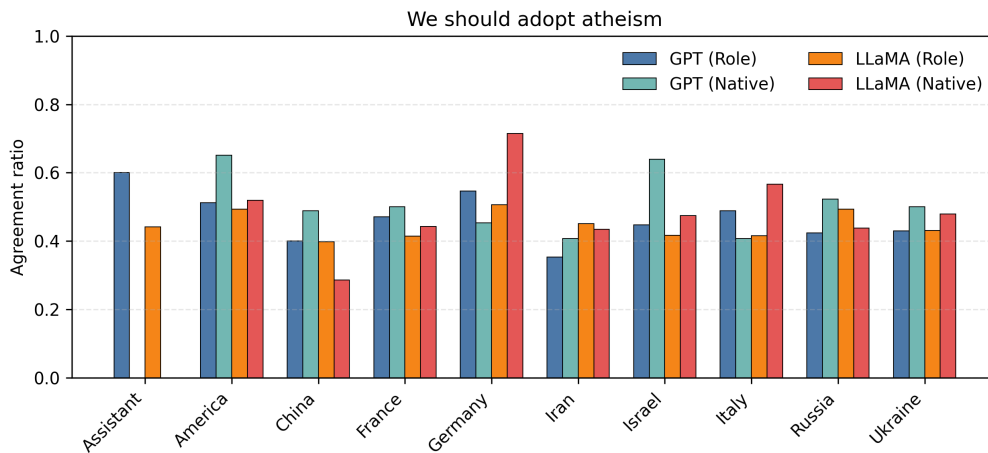


Figure 4.6: Agreement ratios by country for the topic “Adopt atheism.”

Compulsory voting. This topic yields the lowest agreement overall (Figure 4.7). GPT typically sits around 0.2–0.4; LLaMA ranges more widely, with some roles exceeding 0.6. Persona-driven divergence is higher here than in most other topics, again highlighting the role of framing.

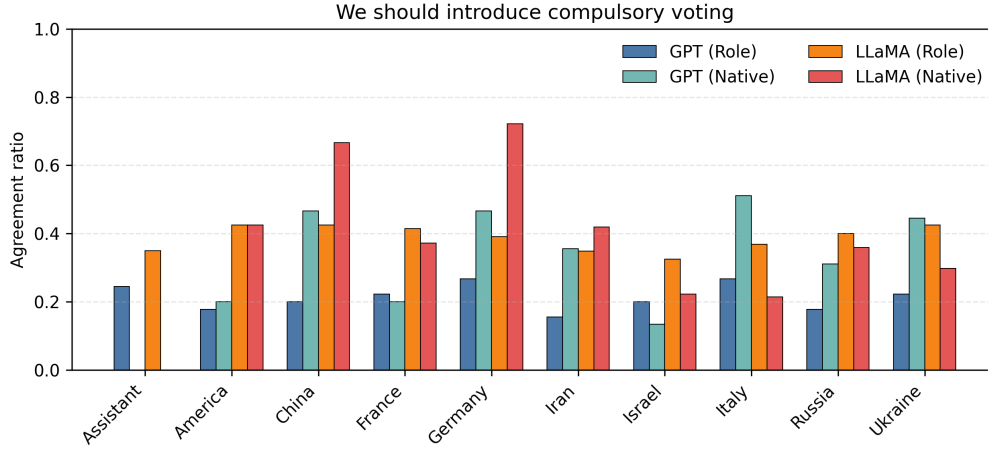


Figure 4.7: Agreement ratios by country for the topic “Compulsory voting.”

Adopt libertarianism. Ratios cluster in the mid-range (0.4–0.7) with clear Role–Native separation in both models, though the size of the gap varies by country (Figure 4.8). The pattern suggests that adopting a role often shifts the model’s stance relative to the Assistant baseline even when overall agreement is moderate.

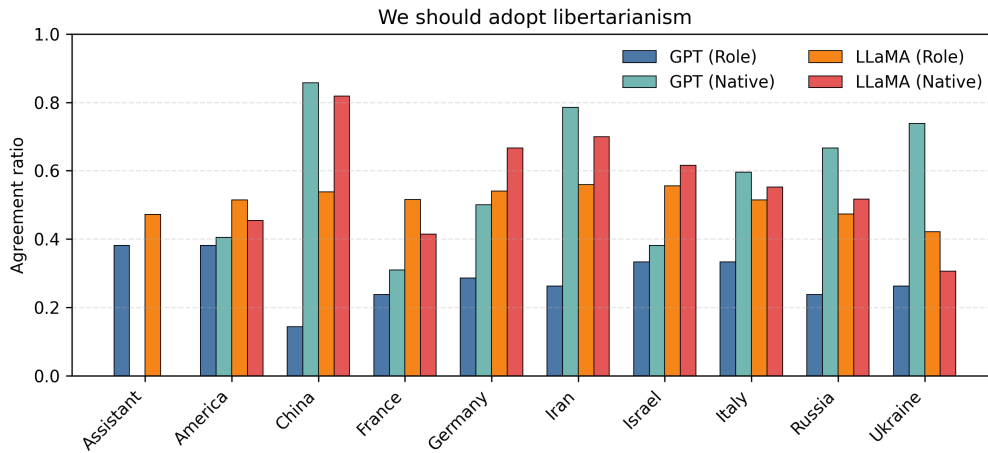


Figure 4.8: Agreement ratios by country for the topic “Adopt libertarianism.”

Summary. Across topics, GPT displays tighter, more stable agreement ratios with smaller Role–Native differences, whereas LLaMA shows larger swings that are strongly topic-dependent. Stability is highest for *multi-party system* (and often for *prostitution*), while the largest persona effects appear in *sanctions*, *atheism*, and *compulsory voting*. The distributional view in Figure 4.9 reinforces this reading: GPT’s spreads are generally narrower, especially under the Role-based and Native conditions, while LLaMA exhibits broader dispersion across topics. The

Assistant baseline remains comparatively steady in both models.

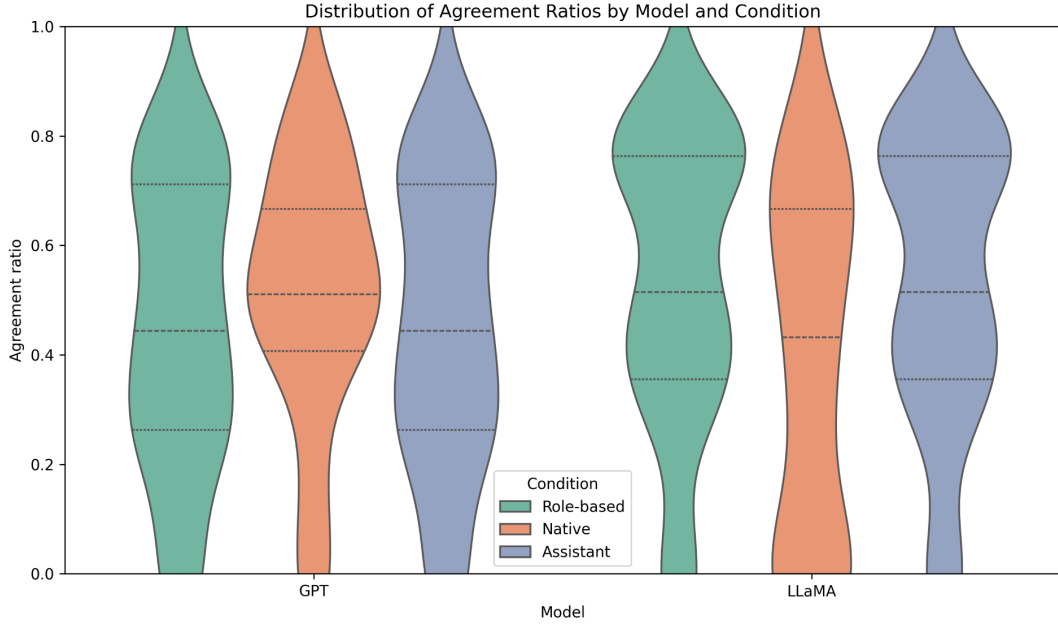


Figure 4.9: Distribution of agreement ratios by model and condition. Each violin shows the spread across topics under *Role-based*, *Native*, and *Assistant* prompts.

4.2 Embedding similarities

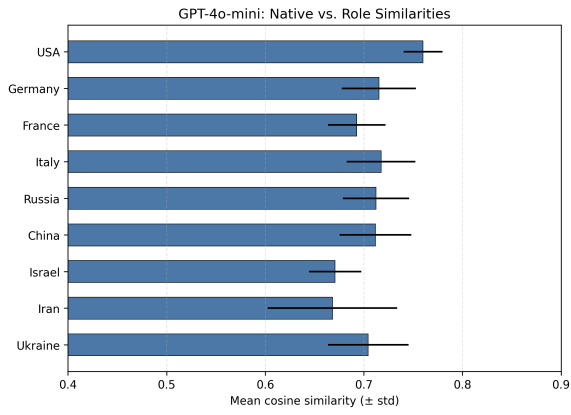
We quantify the semantic proximity between model justifications using mean cosine similarity across the eight topics (whiskers in the bar charts show *standard deviation across topics*, i.e., topic dispersion, not confidence intervals). For each model we report three pairwise comparisons—*Native vs. Role* (NvR), *Role vs. Assistant* (RvA), and *Native vs. Assistant* (NvA)—first as bar charts to capture global patterns by country, then as heatmaps to expose country–topic detail.

4.2.1 GPT-4o-mini

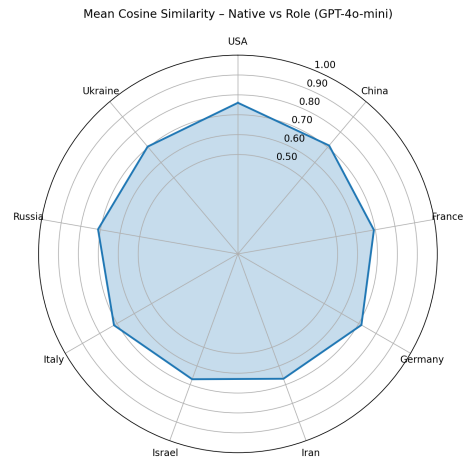
Overall similarity trends. As illustrated in Figure 4.10, *Persona* > *language*. NvA and NvR are consistently higher than RvA for almost all countries (typically by a visible margin), indicating that role prompts reshape reasons more than switching language. NvA and NvR display modest topic dispersion, whereas RvA exhibits the largest dispersion—i.e., stronger topic

dependence—most notably for China and several lower-scoring countries. Country ordering is stable: USA highest; Germany/France/Italy/Russia mid–high; China/Israel/Iran/Ukraine somewhat lower. The radar plots further confirm these tendencies, showing smooth and uniformly high profiles for NvA and NvR across countries, while RvA displays more irregular shapes and lower magnitudes—highlighting greater cross-country variation and reinforcing the stronger influence of persona over language. For a clearer visual comparison, all radar plots are also collected in a single merged figure in Appendix 2a.

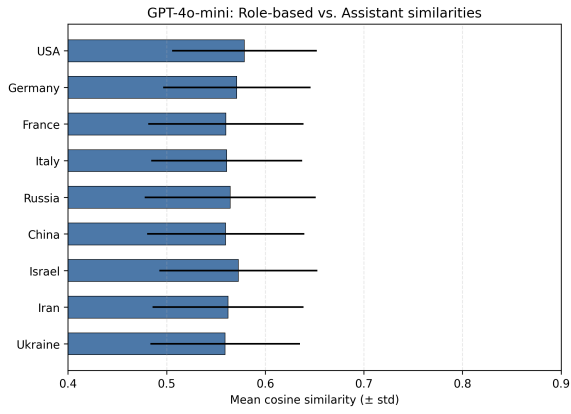
Heatmap observations. As shown in Figure 4.11, across all three comparisons, Multi-party system is among the darkest columns—i.e., highest similarity across countries—and Adopt atheism is also consistently dark. In contrast, No economic sanctions, Compulsory voting, and Adopt libertarianism are repeatedly lighter, especially in the *RvA* heatmap, marking the topics where persona most departs from the assistant baseline. By country, the **USA** tends to remain dark across topics, while **China**—and to a lesser extent **Israel**, **Iran**, and **Ukraine**—shows lighter bands in *RvA* on the low-similarity topics, revealing stronger topic-dependent persona effects.



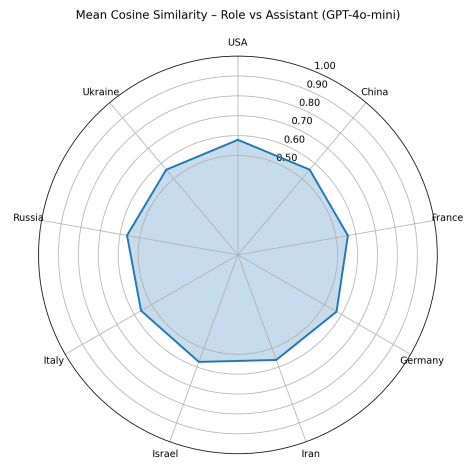
(a) Native vs. Role



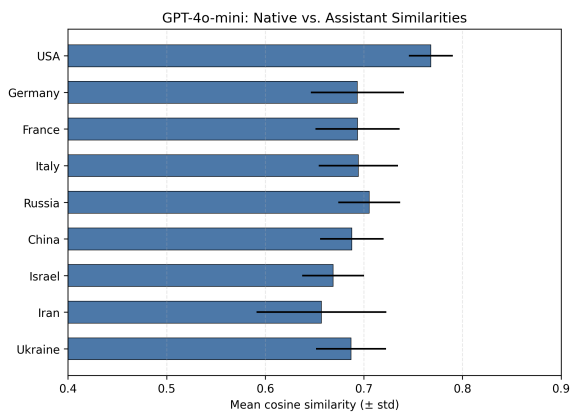
(b) Native vs. Role



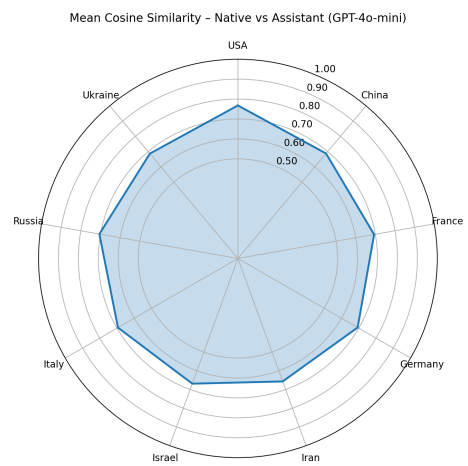
(c) Role vs. Assistant



(d) Role vs. Assistant

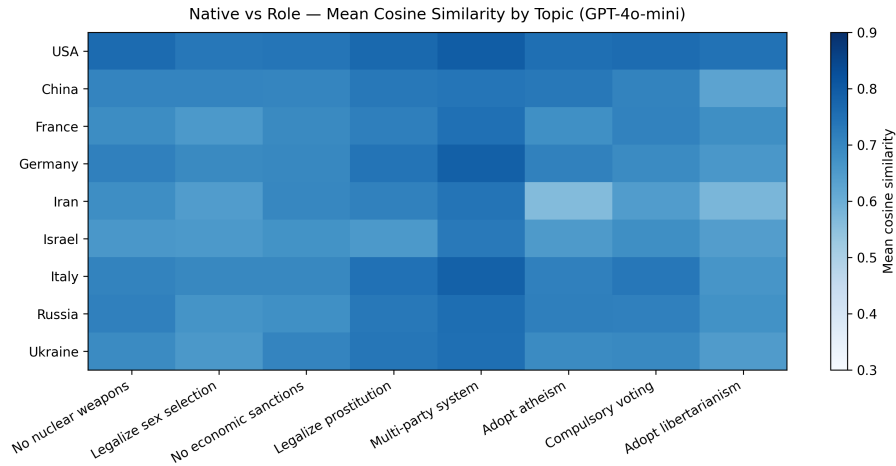


(e) Native vs. Assistant

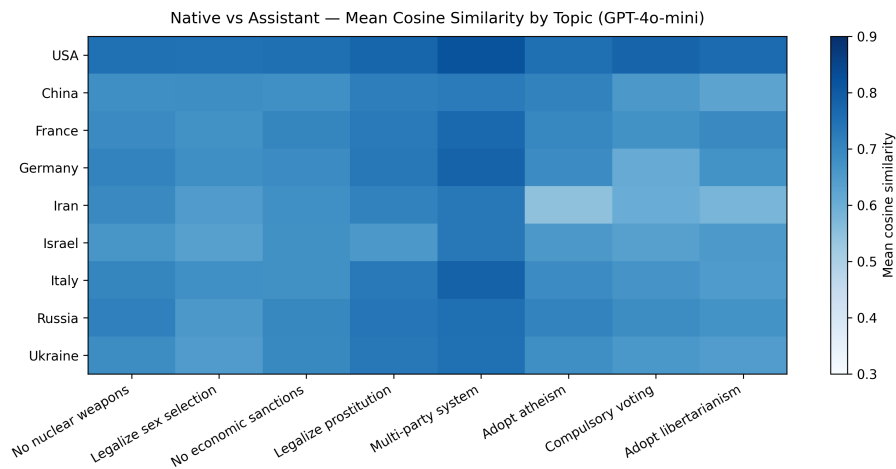


(f) Native vs. Assistant

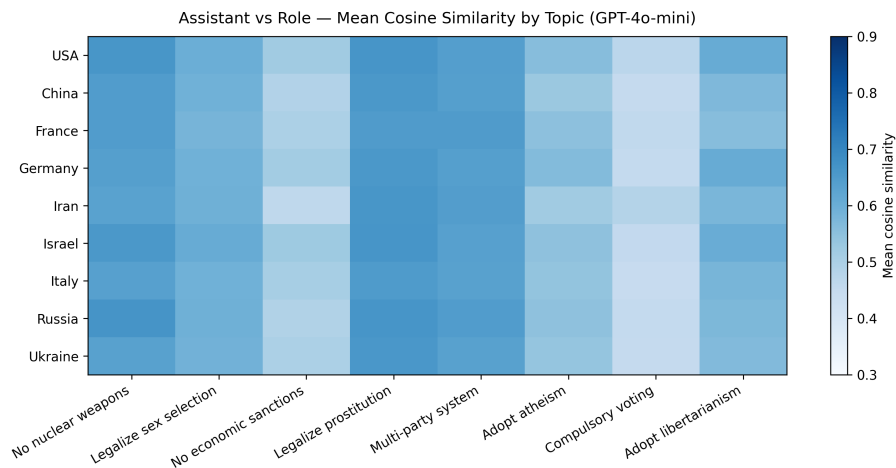
Figure 4.10: GPT-4o-mini — aggregated similarity across conditions.



(a) Native vs. Role (by topic)



(b) Native vs. Assistant (by topic)



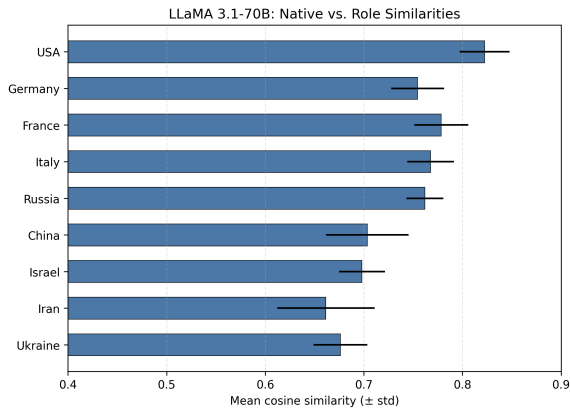
(c) Role vs. Assistant (by topic)

Figure 4.11: **GPT-4o-mini — country-topic heatmaps.** Darker cells indicate higher similarity. Most-stable topics (darker across countries) include *Multi-party system* and *Adopt atheism*. The lowest rows in RvA (lighter cells) frequently occur on *No economic sanctions*, *Compulsory voting*, and *Adopt libertarianism*, revealing where persona prompts diverge most from the assistant baseline.

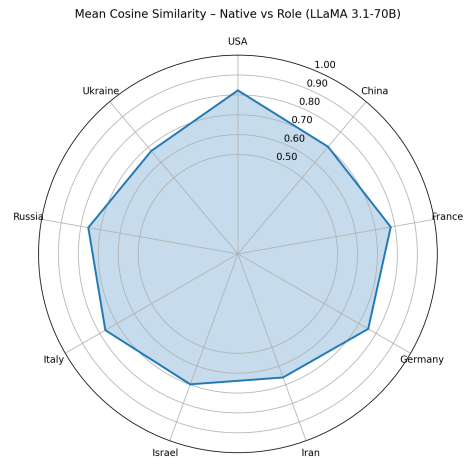
4.2.2 LLaMA 3.1-70B

Overall similarity trends. As shown in Figure 4.12, the qualitative picture mirrors GPT-4o-mini. NvA and NvR are high across countries with modest topic dispersion; RvA is visibly lower with the largest dispersion, again strongest for China. Country structure is stable: USA highest and most consistent; France/Italy/Russia/Germany follow; China/Israel/Iran/Ukraine lower and more topic-sensitive in RvA. The radar plots further reinforce these results, showing consistently high and balanced profiles for NvA and NvR, while RvA exhibits a lower and more irregular shape—confirming stronger cross-country variation and a more pronounced persona effect. For a clearer visual comparison, all radar plots are also collected in a single merged figure in Appendix 2b.

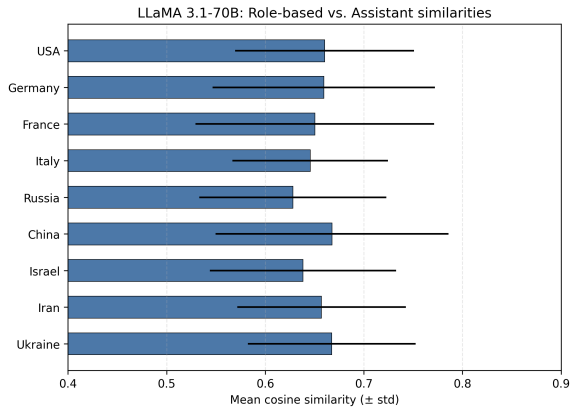
Heatmap observations. As shown in Figure 4.13, Multi-party system (and often Adopt atheism) forms the darkest columns in NvA/NvR, indicating stable, high similarity of reasons. No economic sanctions is notably light in RvA for several countries, with Compulsory voting and Adopt libertarianism also lighter than other topics—these are the main loci of persona-driven divergence from the assistant. Country-wise, the **USA** remains dark across many topics, while **China** shows prominent light bands in RvA on the low-similarity topics; **Israel**, **Iran**, and **Ukraine** also tend to be lighter than the Western European group.



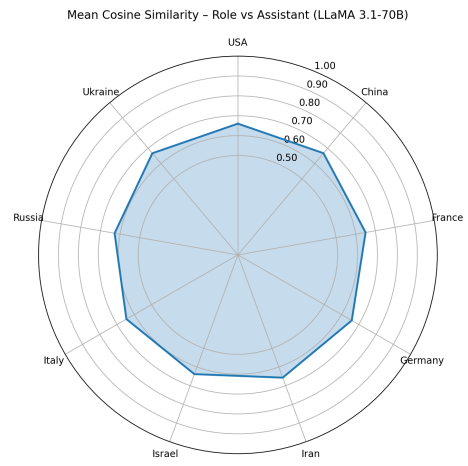
(a) Native vs. Role



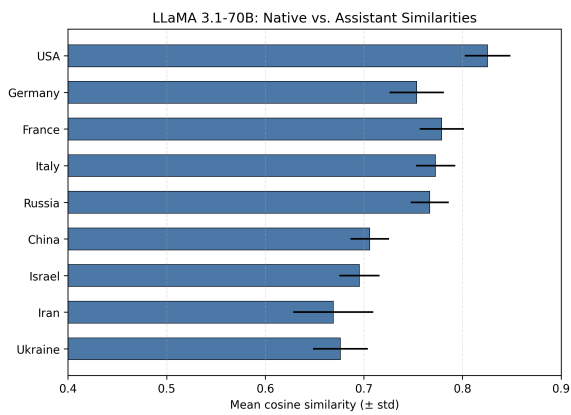
(b) Native vs. Role



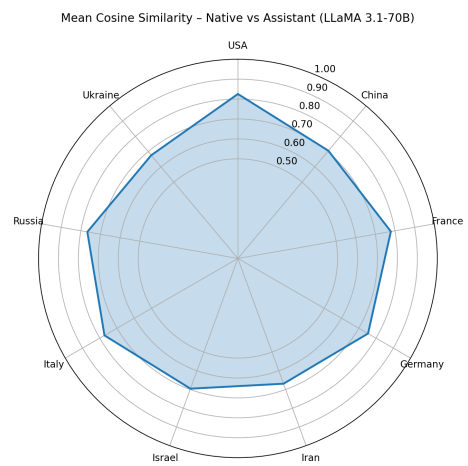
(c) Role vs. Assistant



(d) Role vs. Assistant

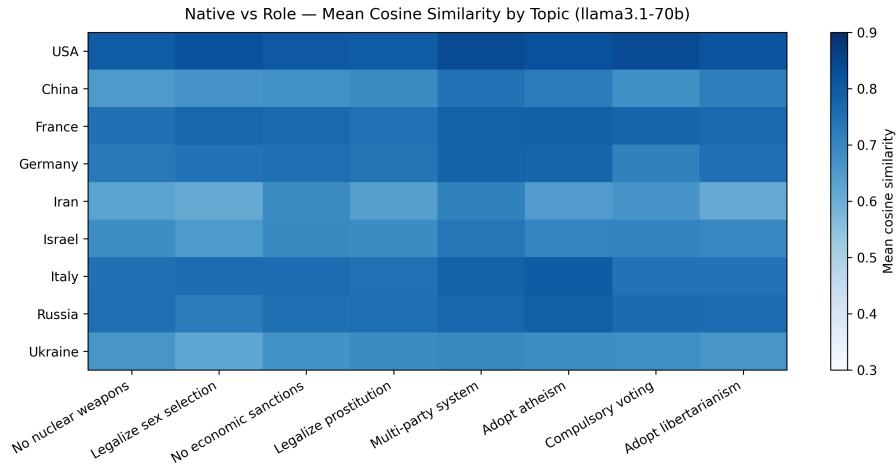


(e) Native vs. Assistant

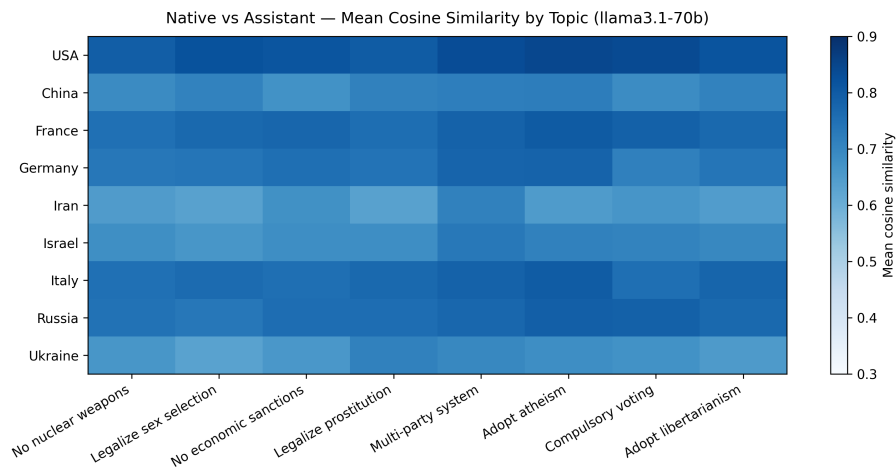


(f) Native vs. Assistant

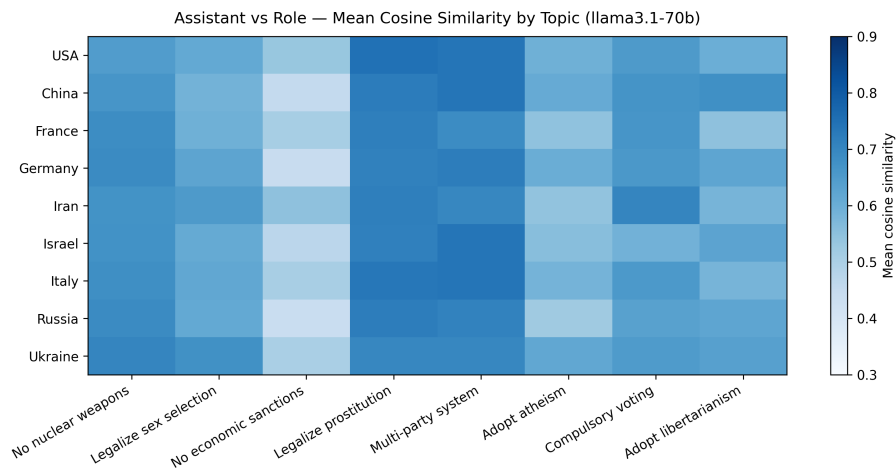
Figure 4.12: LLaMA 3.1-70B — aggregated similarity across conditions.



(a) Native vs. Role (by topic)



(b) Native vs. Assistant (by topic)



(c) Role vs. Assistant (by topic)

Figure 4.13: **LLaMA 3.1-70B — country–topic heatmaps.** Darker = higher similarity. As with GPT-4o-mini, *Multi-party system* and *Adopt atheism* are relatively stable/high across countries in NvA/NvR, whereas RvA is lighter on *No economic sanctions*, *Compulsory voting*, and *Adopt libertarianism*, indicating stronger persona-driven divergence.

4.3 Errors and Neutrality

As shown in Table 4.1, the errors reported here are exclusively due to parsing failures in our pipeline (i.e., cases where the response could not be read as valid JSON). The LLMs themselves successfully generated outputs in all cases, and no generation-level errors were observed.

GPT models did not produce neutral answers in any condition (all neutrality rates are 0). Across the three GPT conditions (Assistant, Native, Role), there was only a single error case in total (Native–GPT: 1/5949, $\approx 0.017\%$), caused by a response that failed JSON parsing. In contrast, LLaMA showed higher error rates (Assistant–LLaMA: 779/5949, 13.0%; Role–LLaMA: 871/5949, 14.6%; Native–LLaMA: 863/5949, 14.5%) and a small but non-zero neutrality rate only in the Native condition (74/5949, 1.2%).

Condition–Model	Error rate	Neutrality rate	Error count	Neutral count
Assistant – GPT	0.000	0.000	0	0
Assistant – LLaMA	0.130	0.000	779	0
Native – GPT	0.000	0.000	1	0
Native – LLaMA	0.145	0.012	863	74
Role – GPT	0.000	0.000	0	0
Role – LLaMA	0.146	0.000	871	0

Table 4.1: Error and neutrality summary across condition–model pairs. Error rate is computed as *error_count/total_rows*; neutrality rate as *neutral_count/total_rows*.

Chapter 5

Conclusions

This study set out to examine ideological bias in large language models by looking not only at how often they agree or disagree, but at how they justify those stances across topics, roles, and languages. As we moved across the nine topics, a pattern emerged: some questions seemed to invite a shared framing that held steady no matter how we prompted the models, while others fractured as soon as we shifted the persona. *Multi-party system*—and often *Adopt atheism*—tended to hold together: justifications clustered closely across conditions, as if the models drew on a relatively uniform background narrative. By contrast, *No economic sanctions*, *Compulsory voting*, and *Adopt libertarianism* repeatedly came apart when we introduced national roles; the same prompt in a different voice tugged the reasoning in noticeably different directions. In other words, variation was less a constant “property of the model” and more a topic-sensitive response to how the question was framed.

Persona cues amplified this effect far more than language did. When we compared *Role vs. Assistant* (RvA), divergences in the reasons were consistently larger than either *Native vs. Assistant* (NvA) or *Native vs. Role* (NvR). Switching languages mostly modulated tone and emphasis; assigning a country-specific role re-shaped the stance and its rationale. This held across models, though not equally: GPT-4o-mini was generally steadier, while LLaMA 3.1-70B swung more when the persona changed, suggesting that post-training alignment may dampen these shifts in GPT while leaving LLaMA somewhat freer to follow the role cue. A similar geography appeared across national roles. In our runs, the USA often sat at the high and stable

end of the spectrum, Western European countries such as France, Italy, and Germany clustered in a mid-high band, while China, Israel, Iran, and Ukraine tended to be lower and more topic-sensitive—patterns that read like echoes of the cultural framings present in the training data.

Taken together, the two lenses—agreement ratios and embedding-based similarity—tell a complementary story. Agreement counts show where stances line up; the embeddings reveal that even when the surface verdicts converge, the underlying reasons can drift when a persona is applied. For applications that trade on neutrality—journalism, policy analysis, education—this matters: a model may sound fluent in many languages yet deliver role-conditioned reasoning that shifts its ideological center. The upshot is practical as much as theoretical: bias evaluation should remain systematic and transparent, with persona prompts treated as first-class stress tests rather than incidental details, and with continuous monitoring to track how alignment changes the balance between stability and sensitivity over time.

Chapter 6

Limitations and Future works

While the analysis offered useful insights, it also comes with certain limitations. First, the study was restricted to a limited set of eight countries and their native languages, which narrows the cultural and linguistic diversity of the results. Second, the scope of ideological topics was bound by the coverage of the chosen dataset, meaning that several relevant areas of debate were left unexplored. Finally, no external ground-truth source for ideological alignment was used, so validation relied solely on comparisons across models and conditions rather than on independent reference points.

These limitations point directly to future directions. A natural extension is to broaden the set of countries and languages, which would allow for a more comprehensive cross-cultural evaluation. Future work could also compare model outputs against external reference sources such as Wikipedia or curated expert-written texts, which would serve as a more objective anchor for assessing ideological stance. Another direction is the use of richer and better-labeled datasets that span a wider range of ideological topics, making it possible to test model behavior beyond the current domains. It would also be valuable to explore repeated prompting of identical arguments to evaluate intra-model variance—that is, whether the same model gives consistent answers to the same question, or whether subtle stochastic effects cause ideological drift. Finally, it is worth considering whether the location from which an LLM is accessed influences its behavior, for example through region-specific filtering, safety layers, or hidden geolocation conditioning. Such factors could further affect the reproducibility and neutrality of model outputs in different

contexts.

Overall, these directions suggest that while the present study highlights systematic patterns of ideological bias, there is considerable room to expand the analysis, validate the findings against external standards, and account for geographic and contextual factors that may shape LLM behavior in practice.

References

- [1] OpenAI et al. *GPT-4 Technical Report*. 2023. arXiv: 2303.08774 [cs.CL]. URL: <https://arxiv.org/abs/2303.08774>.
- [2] Hugo Touvron et al. *LLaMA: Open and Efficient Foundation Language Models*. 2023. arXiv: 2302.13971 [cs.CL]. URL: <https://arxiv.org/abs/2302.13971>.
- [3] Laura Weidinger et al. *Ethical and Social Risks of Harm from Language Models*. 2021. arXiv: 2112.04359 [cs.CL]. URL: <https://arxiv.org/abs/2112.04359>.
- [4] Shibani Santurkar et al. “Whose Opinions Do Language Models Reflect?” In: *Proceedings of the 40th International Conference on Machine Learning*. Ed. by Andreas Krause et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 29971–30004. URL: <https://proceedings.mlr.press/v202/santurkar23a.html>.
- [5] Junfeng Jiao et al. *Navigating LLM Ethics: Advancements, Challenges, and Future Directions*. 2025. arXiv: 2406.18841 [cs.CY]. URL: <https://arxiv.org/abs/2406.18841>.
- [6] Keita Kurita et al. *Measuring Bias in Contextualized Word Representations*. 2019. arXiv: 1906.07337 [cs.CL]. URL: <https://arxiv.org/abs/1906.07337>.
- [7] Moin Nadeem, Anna Bethke, and Siva Reddy. *StereoSet: Measuring stereotypical bias in pretrained language models*. 2020. arXiv: 2004.09456 [cs.CL]. URL: <https://arxiv.org/abs/2004.09456>.
- [8] Md Abdul Aowal et al. *Detecting Natural Language Biases with Prompt-based Learning*. 2023. arXiv: 2309.05227 [cs.CL]. URL: <https://arxiv.org/abs/2309.05227>.

- [9] Nicole Gross. “What ChatGPT Tells Us about Gender: A Cautionary Tale about Performativity and Gender Biases in AI”. In: *Social Sciences* 12.8 (2023). ISSN: 2076-0760. DOI: 10.3390/socsci12080435. URL: <https://www.mdpi.com/2076-0760/12/8/435>.
- [10] Aleksandra Urman and Mykola Makhortykh. “The silence of the LLMs: Cross-lingual analysis of guardrail-related political bias and false information prevalence in ChatGPT, Google Bard (Gemini), and Bing Chat”. In: *Telematics and Informatics* 96 (2025), p. 102211. ISSN: 0736-5853. DOI: <https://doi.org/10.1016/j.tele.2024.102211>. URL: <https://www.sciencedirect.com/science/article/pii/S0736585324001151>.
- [11] Pranav Narayanan Venkit et al. *Nationality Bias in Text Generation*. 2023. arXiv: 2302.02463 [cs.CL]. URL: <https://arxiv.org/abs/2302.02463>.
- [12] Richard Rogers and Xiaoke Zhang. “The Russia–Ukraine War in Chinese Social Media: LLM Analysis Yields a Bias Toward Neutrality”. In: *Social Media + Society* 10 (May 2024). DOI: 10.1177/20563051241254379.
- [13] Tomas Mikolov et al. *Efficient Estimation of Word Representations in Vector Space*. 2013. arXiv: 1301.3781 [cs.CL]. URL: <https://arxiv.org/abs/1301.3781>.
- [14] Jeffrey Pennington, Richard Socher, and Christopher Manning. “GloVe: Global Vectors for Word Representation”. In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Ed. by Alessandro Moschitti, Bo Pang, and Walter Daelemans. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1532–1543. DOI: 10.3115/v1/D14-1162. URL: <https://aclanthology.org/D14-1162/>.
- [15] Matthew E. Peters et al. *Deep contextualized word representations*. 2018. arXiv: 1802.05365 [cs.CL]. URL: <https://arxiv.org/abs/1802.05365>.
- [16] Jacob Devlin et al. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. 2019. arXiv: 1810.04805 [cs.CL]. URL: <https://arxiv.org/abs/1810.04805>.

- [17] Nils Reimers and Iryna Gurevych. *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*. 2019. arXiv: 1908.10084 [cs.CL]. URL: <https://arxiv.org/abs/1908.10084>.
- [18] Fangxiaoyu Feng et al. *Language-agnostic BERT Sentence Embedding*. 2022. arXiv: 2007.01852 [cs.CL]. URL: <https://arxiv.org/abs/2007.01852>.
- [19] Ashish Vaswani et al. *Attention Is All You Need*. 2023. arXiv: 1706.03762 [cs.CL]. URL: <https://arxiv.org/abs/1706.03762>.
- [20] Alec Radford et al. *Improving language understanding by generative pre-training*. 2018.
- [21] Tom B. Brown et al. *Language Models are Few-Shot Learners*. 2020. arXiv: 2005.14165 [cs.CL]. URL: <https://arxiv.org/abs/2005.14165>.
- [22] Long Ouyang et al. *Training language models to follow instructions with human feedback*. 2022. arXiv: 2203.02155 [cs.CL]. URL: <https://arxiv.org/abs/2203.02155>.
- [23] Bernd Carsten Stahl and Damian Eke. *The ethics of ChatGPT – Exploring the ethical issues of an emerging technology*. 2024. DOI: <https://doi.org/10.1016/j.ijinfomgt.2023.102700>. URL: <https://www.sciencedirect.com/science/article/pii/S0268401223000816>.
- [24] Nikita Nangia et al. *CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models*. 2020. arXiv: 2010.00133 [cs.CL]. URL: <https://arxiv.org/abs/2010.00133>.
- [25] Patrick Haller, Ansar Aynedinov, and Alan Akbik. *OpinionGPT: Modelling Explicit Biases in Instruction-Tuned LLMs*. 2023. arXiv: 2309.03876 [cs.CL]. URL: <https://arxiv.org/abs/2309.03876>.
- [26] Anthony G. Greenwald, Debbie E. McGhee, and Jordan L. K. Schwartz. “Measuring Individual Differences in Implicit Cognition: The Implicit Association Test”. In: *Journal of Personality and Social Psychology* 74.6 (1998), pp. 1464–1480. DOI: 10.1037/0022-3514.74.6.1464. URL: <https://psycnet.apa.org/doi/10.1037/0022-3514.74.6.1464>.

- [27] Chandler May et al. *On Measuring Social Biases in Sentence Encoders*. 2019. arXiv: 1903.10561 [cs.CL]. URL: <https://arxiv.org/abs/1903.10561>.
- [28] Nailia Mirzakhmedova et al. *The Touché23-ValueEval Dataset for Identifying Human Values behind Arguments*. 2023. arXiv: 2301.13771 [cs.CL]. URL: <https://arxiv.org/abs/2301.13771>.
- [29] Nikita Soni et al. “Human Language Modeling”. In: *Findings of the Association for Computational Linguistics: ACL 2022*. Ed. by Smaranda Muresan, Preslav Nakov, and Aline Villavicencio. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 622–636. DOI: 10.18653/v1/2022.findings-acl.52. URL: <https://aclanthology.org/2022.findings-acl.52/>.
- [30] Maarten Grootendorst. *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*. 2022. arXiv: 2203.05794 [cs.CL]. URL: <https://arxiv.org/abs/2203.05794>.
- [31] Nidhal Baccouri. *deep-translator: A flexible free and unlimited python tool to translate between different languages in a simple way using multiple translators*. <https://pypi.org/project/deep-translator/>. Version 1.11.4, accessed: 2025-08-24. 2020.
- [32] OpenAI. *Learn how to use JSON mode*. <https://platform.openai.com/docs/guides/json-mode>. Accessed: 2025-07-02. 2025.

Appendix

Translation Maps

The following table summarizes the mapping of stance expressions across different languages.

Each language has three categories: *Agree*, *Disagree*, and *Neutral*.

Table 1: Translation map of stance expressions across languages.

Language	Agree	Disagree	Neutral
English	agree	disagree	–
Italian	d'accordo, sì, sono d'accordo, sono d'accordo ma con delle precisazioni, condivido, rispetto, concordo, in parte d'accordo, parzialmente d'accordo, non sono completamente d'accordo, non del tutto d'accordo	non sono d'accordo, in disaccordo, discordo, contro, parzialmente in disaccordo, parziale disaccordo, non sono in grado di esprimere un'opinione personale	ambiguo, interessante, è una questione complessa, è un argomento complesso, complessa, dubbi, discutibile
French	d'accord, je suis d'accord, plutôt d'accord, je suis d'accord en partie, je suis d'accord partiellement, d'accord avec certaines réserves, d'accord avec des réserves	pas d'accord, désaccord, en désaccord, disaccord, non, je ne suis pas d'accord, je ne suis pas entièrement d'accord, je ne suis pas complètement d'accord, je ne suis pas sûr, je suis en désaccord, je suis partiellement en désaccord, je suis en désaccord partiel, pas nécessairement d'accord	nuancé, neutre, discutable, difficile à dire, je vois les deux côtés, je vois les deux côtés de la question, je comprends, je comprends les arguments éthiques et religieux, je vois votre point de vue, je suis partagé

Continued on next page

Table 1 – Continued from previous page

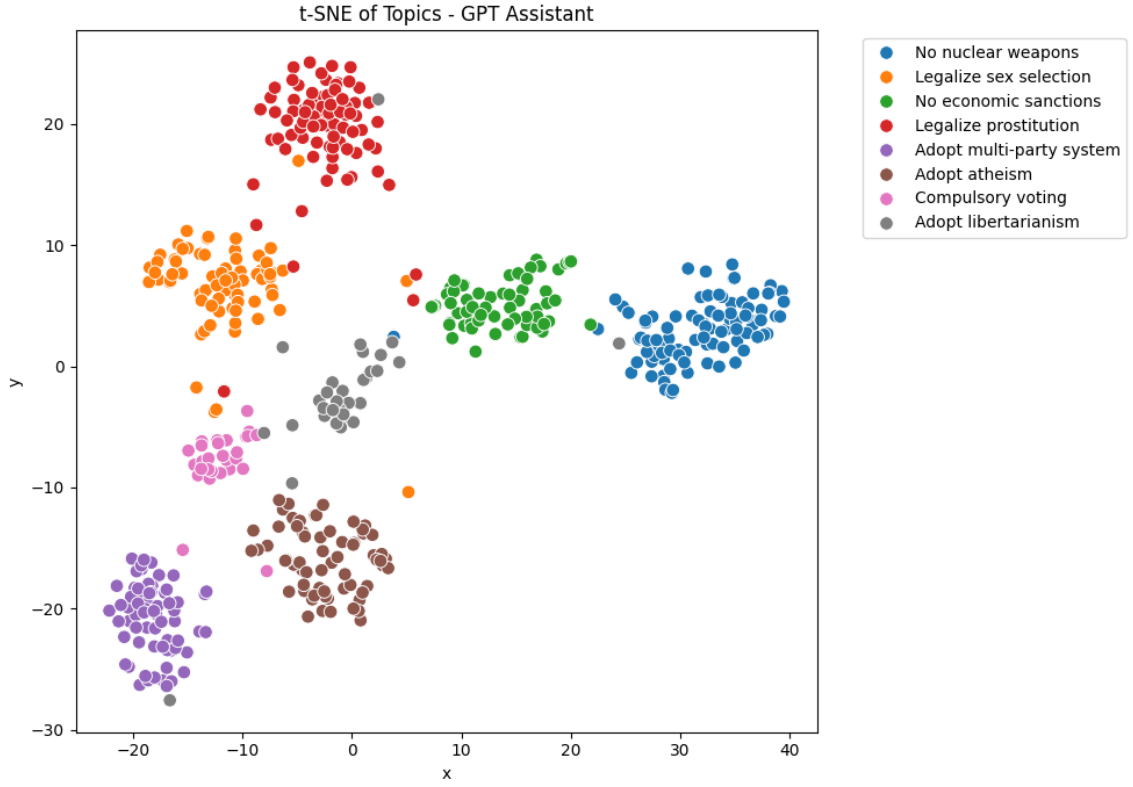
Language	Agree	Disagree	Neutral
Chinese	同意, 支持, 认同, 赞同, 我同意, 是的, 对	不同意, 反对, 否定, 否决, 不完全赞同, 不完全同意, 不完全正确, 我不同意, 不, 反对核武器, 卖淫与许多强奸、骚扰和 谋杀有关	部分同意, 部分正确, 部分, 不一定, 有争议, 批判性, 担忧, 人口失衡可能会 导致社会、经济和文化问题, 宗教信仰多样化, 自由主义
German	stimme zu, zustimmung, ja, ja ich stimme zu, ja stimme zu, ja das stimmt, ja das ist richtig	stimme nicht zu, nicht stimmen, stimmen sie nicht zu, stimmen nicht zu, stimmen gegen, stimme gegen, nein, ich stimme nicht zu, nicht gut	neutral, problematisch, bedenklich, interessant, komplex, es ist zweifelhaft, interessant aber komplex
Russian	согласен, соглашусь, согласен частично, в определенной степени согласен	не согласен, несогласен, не полностью согласен, не вполне согласен, нет, против, не совсем согласен, не совсем	не уверен, возможно, сомневаюсь, нейтрально, комплексный вопрос, неоднозначен, соглашусь с оговорками
Ukrainian	погодитися, згодитися, підтверджую, підтримую, так, точно	не погодитися, відхиляю, ні, відмовитися, відхиляти, непогодитися, відсутність згоди	відповідь залежить від багатьох чинників, залежить від ситуації, немає чіткої відповіді, відкрите питання, більш-менш згоден

Continued on next page

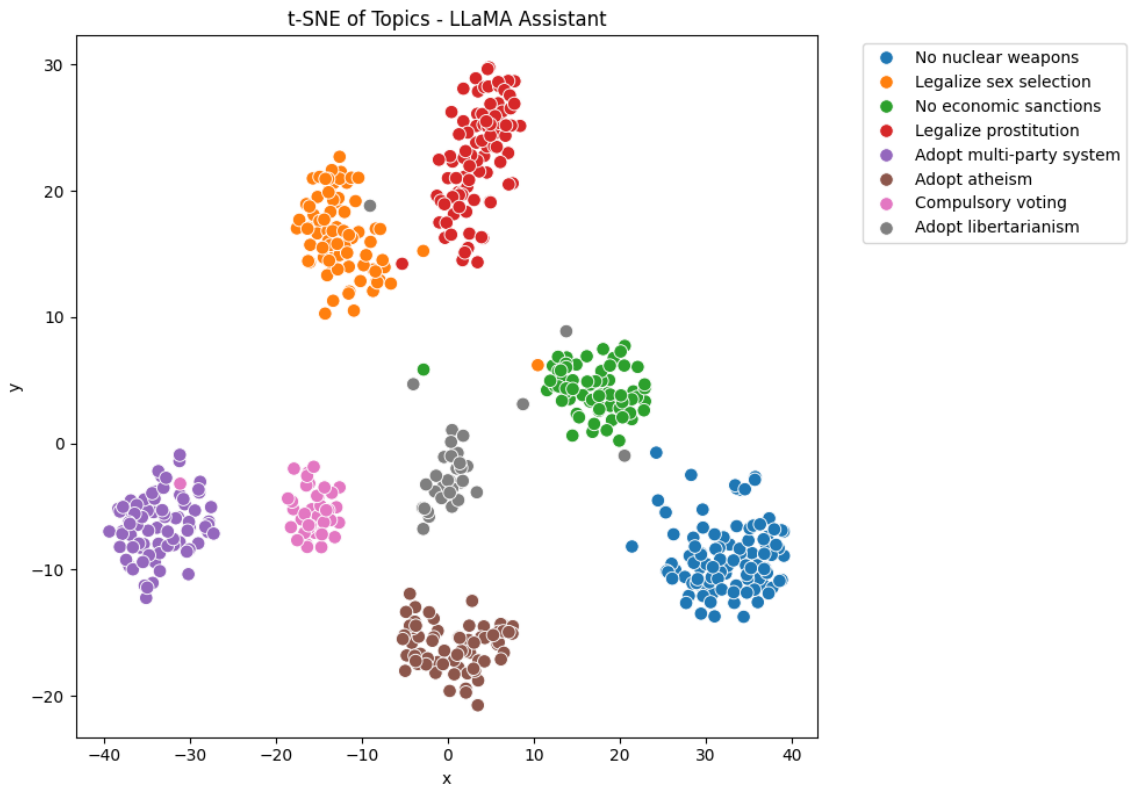
Table 1 – *Continued from previous page*

Language	Agree	Disagree	Neutral
Persian	موافق	نیستم غیرموافق, مخالف, خلاف منفی, موافق,	خنثی
Hebrew	מסכים חלקית מסכים	מסכים איני מסכים לא מסכים איננו	חלוקה הסכמה אין חלוקה

t-SNE Projections of Topic Embeddings across Models and Conditions

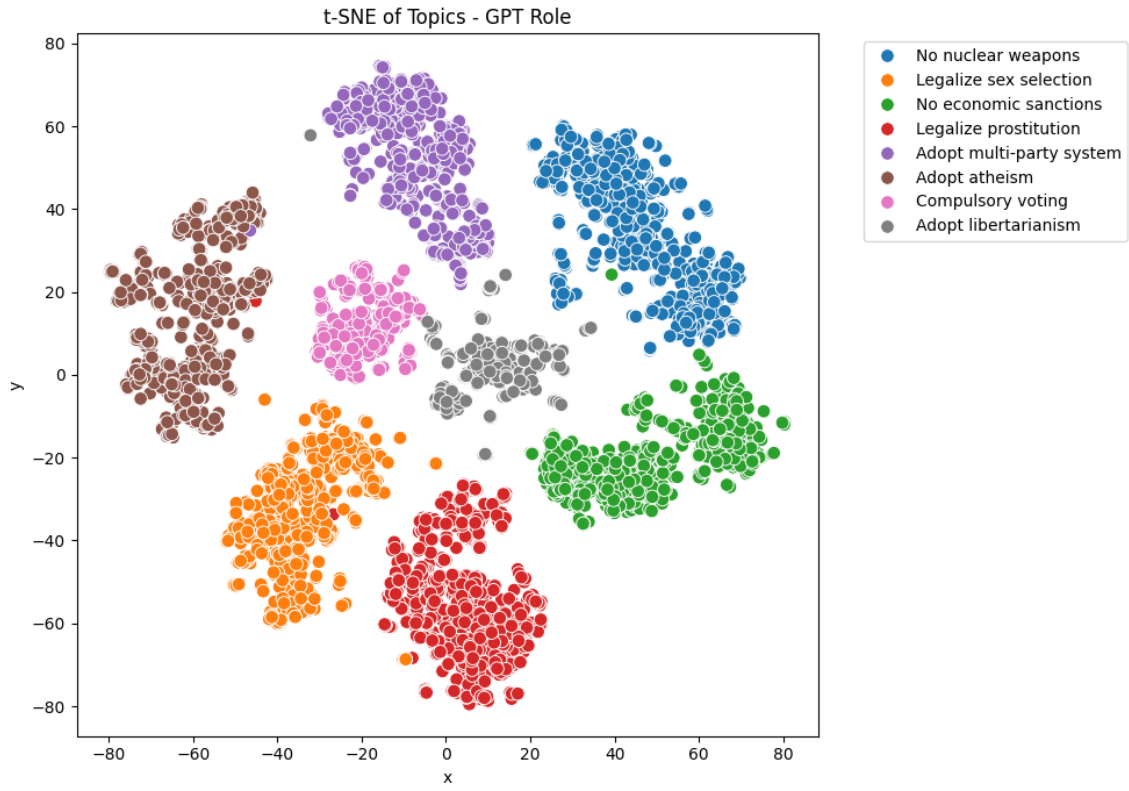


(a) GPT — Assistant

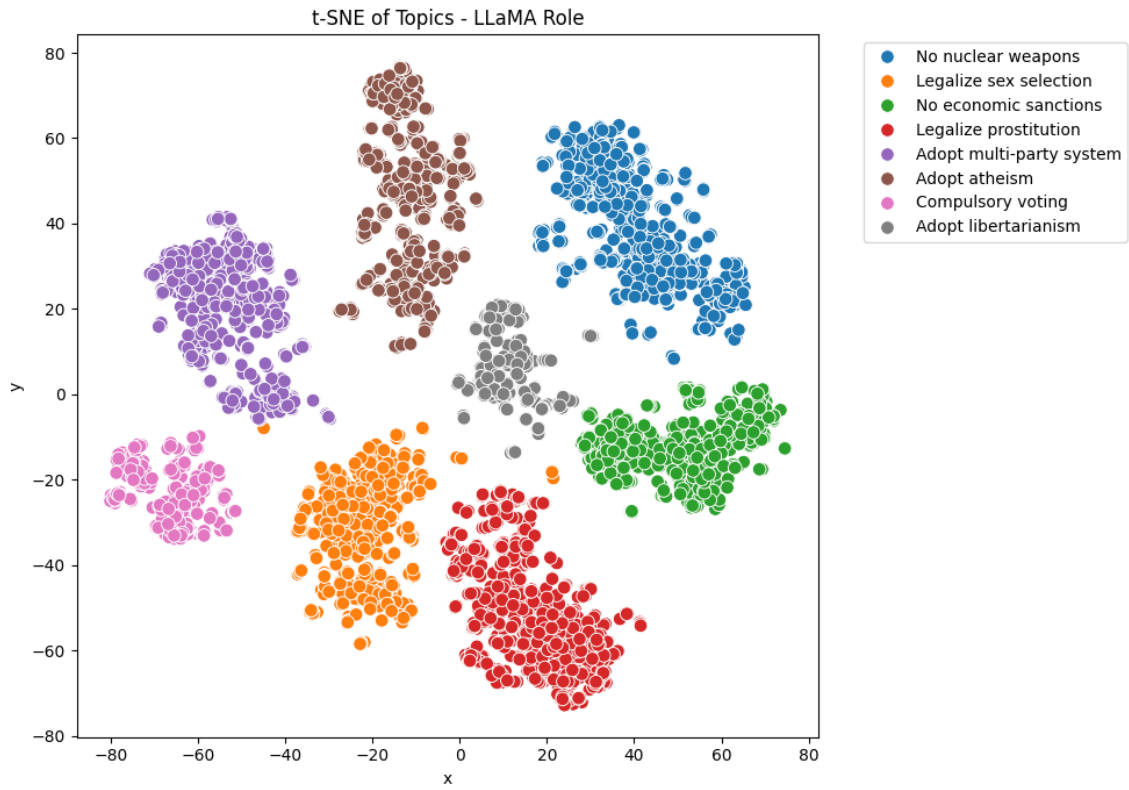


(b) LLaMA — Assistant

Figure 1: t-SNE projections of reason embeddings by topic for GPT and LLaMA across conditions.

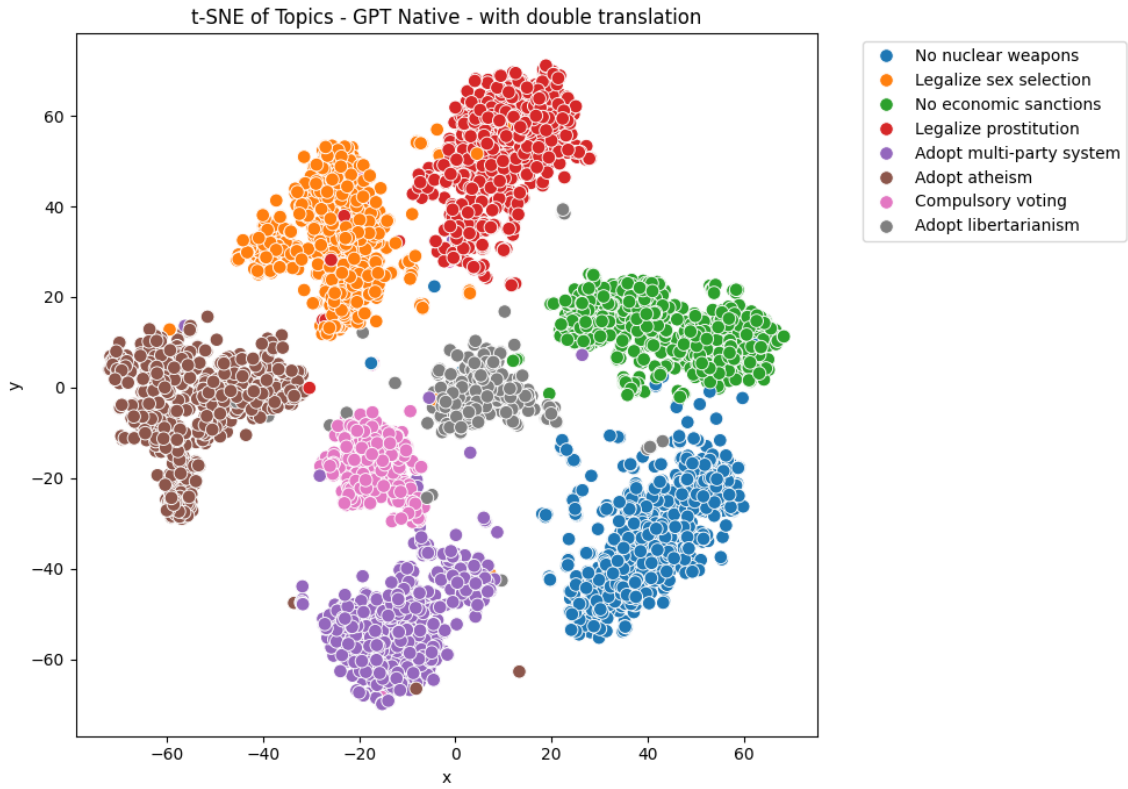


(c) GPT — Role

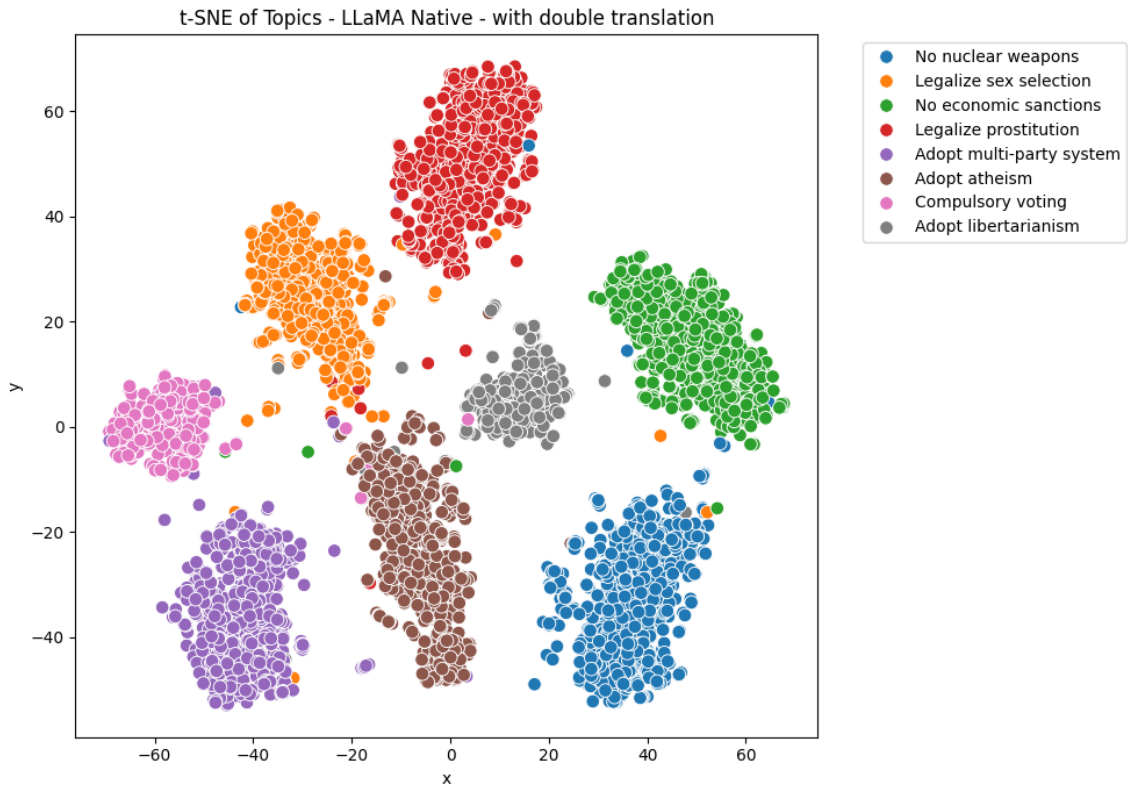


(d) LLaMA — Role

Figure 1: t-SNE projections (continued: Role).

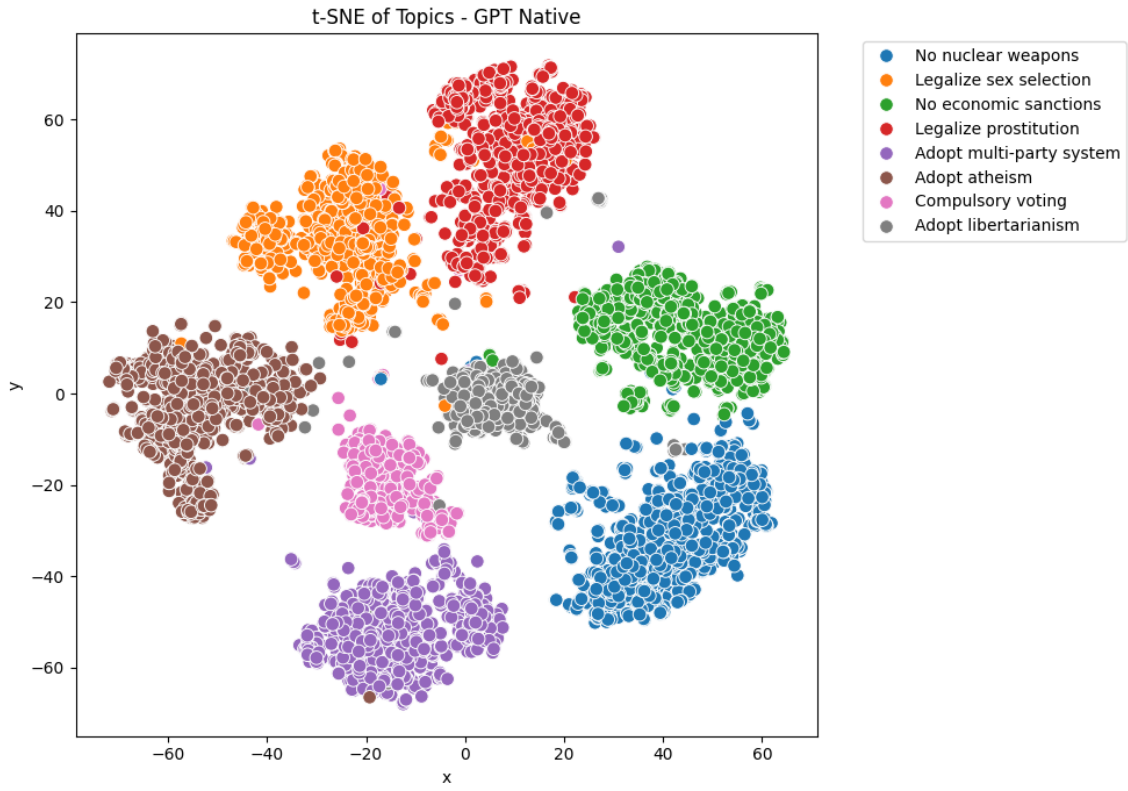


(e) GPT — Native (double translation)

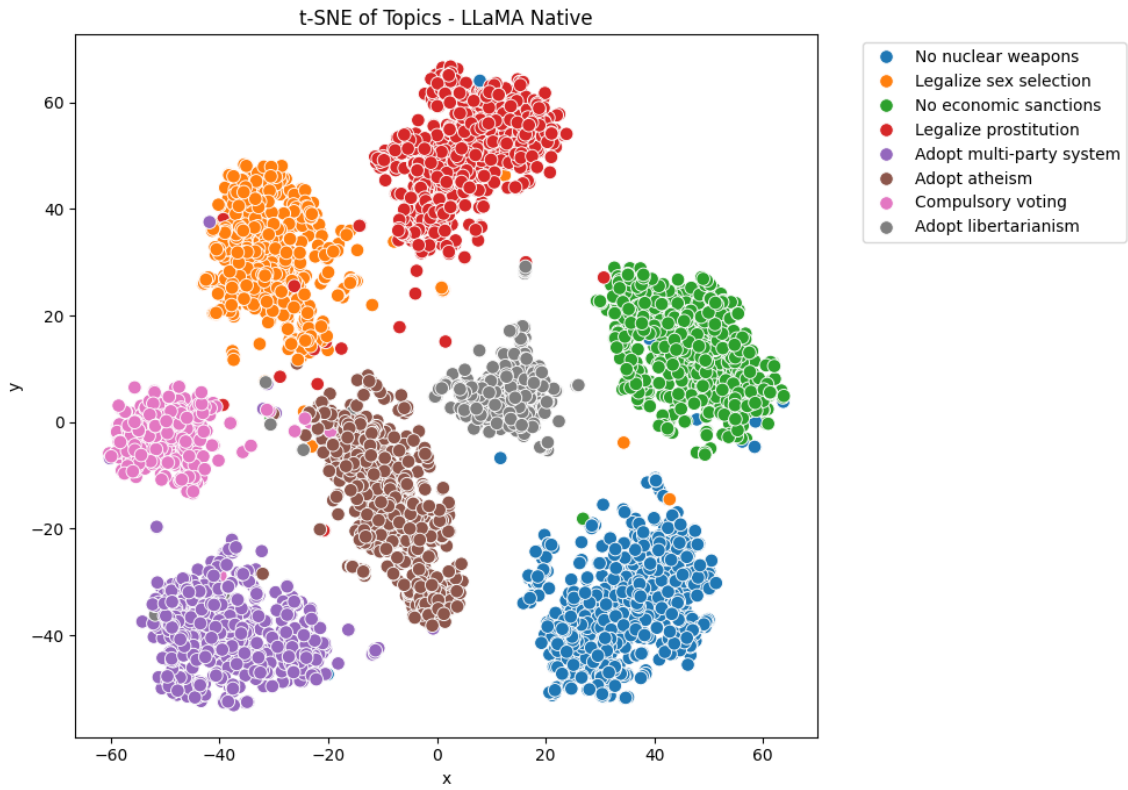


(f) LLaMA — Native (double translation)

Figure 1: t-SNE projections (continued: Native—double translation).



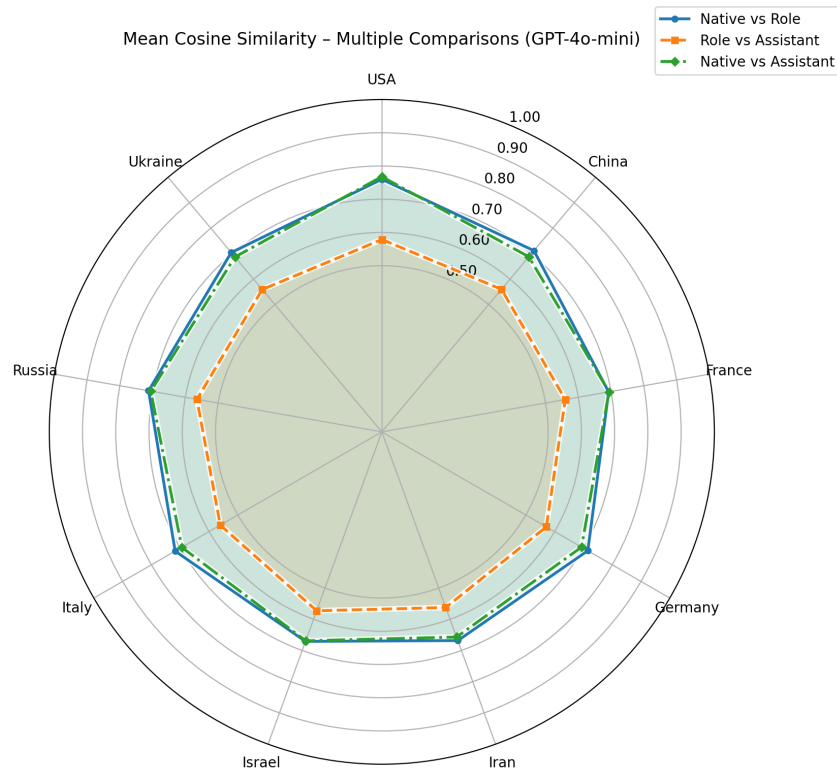
(g) GPT — Native



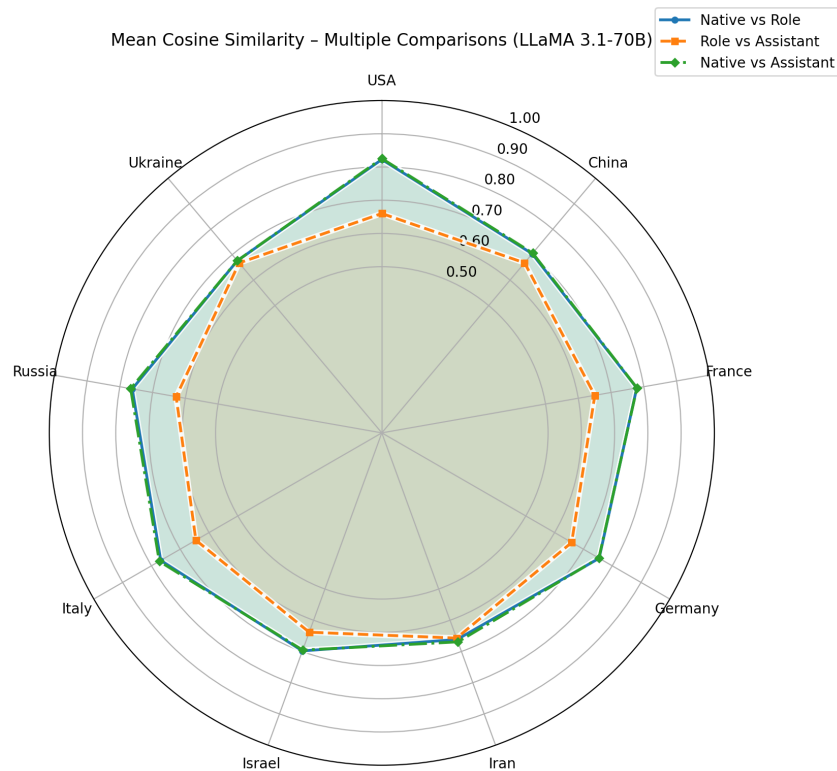
(h) LLaMA — Native

Figure 1: t-SNE projections (continued).

Aggregated Radar Plots for Cross-Condition Comparison



(a) GPT-4o-mini — merged radar plots (NvR, RvA, NvA).



(b) LLaMA 3.1-70B — merged radar plots (NvR, RvA, NvA).

Figure 2: Merged radar plots for both models.